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Thesis

WHAT DOES A PHYSICS NETWORK ACTUALLY LEARN? MECHANISTIC INTERPRETABILITY ACROSS A SPECTRUM OF PHYSICS-INFORMED NEURAL ARCHITECTURES

by

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B.S., **[TODO: Institution]**, **[TODO: Year]**

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ABSTRACT

[Write last. One paragraph each on the problem, the method, the results, and the conclusions [BU 1.5.1].]

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[Fills in automatically once tables exist.]

LIST OF FIGURES

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LIST OF ABBREVIATIONS

CKA	Centered Kernel Alignment
EGNN	E(n) Equivariant Graph Neural Network
GNN	Graph Neural Network
HNN	Hamiltonian Neural Network
MLP	Multilayer Perceptron
MSE	Mean Squared Error
RQ	Research Question
SO(3)	Special Orthogonal Group in Three Dimensions (rotations)

CHAPTER ONE

INTRODUCTION

[Motivates the problem, poses the central question, states the hypothesis and the five research questions, and lists the intended contributions. Adapted from the proposal's Problem Statement and Proposed Research; Contributions and Organization are written last.]

1.1 Motivation

1.2 The Central Question

1.3 Why Coulomb Dynamics

1.4 Thesis Statement

1.5 Research Questions

1.6 Contributions

1.7 Thesis Organization

CHAPTER TWO

BACKGROUND AND RELATED WORK

[Everything the reader needs to follow Chapters 3–8: the spectrum of physics priors, each architecture family, the interpretability tools, and the closest prior work (Cranmer et al. 2020 in particular). Related-work survey still in progress.]

2.1 Learning Physical Dynamics with Neural Networks

2.2 A Spectrum of Physics Priors

2.3 Symmetry-Constrained Architectures

2.3.1 Equivariance

2.3.2 Equivariant Graph Neural Networks

2.3.3 Attention-Based Equivariant Message Passing

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2.6.2 Causal Intervention

2.6.3 Representational Similarity

2.6.4 Grokking and Delayed Generalization

2.7 Interpretability of Physics and Scientific ML Models

2.8 Summary and Gap

CHAPTER THREE

THE COULOMB DYNAMICS TESTBED

[Fully specifies the physical system, simulator, verification checks, prediction task, dataset, and the analytic ground-truth quantities that serve as probe targets (the “answer key”). Decision pending: single-step acceleration vs. multi-step position prediction.]

3.1 Physical System

3.2 Simulator

3.3 Physics Verification

3.4 Prediction Task

3.5 Dataset

3.6 Ground-Truth Physical Quantities

CHAPTER FOUR

ARCHITECTURES AND TRAINING

[Defines the architecture suite, per-model implementation details (including where "layer k activations" are read from), the matched training protocol, and shows comparable output accuracy across models so that later representational differences are not explained by one model being worse. Decision pending: five vs. six architectures.]

4.1 Architecture Suite

4.2 Implementation Details

4.2.1 Multilayer Perceptron

4.2.2 Non-Equivariant Graph Neural Network

4.2.3 EGNN

4.2.4 Attention EGNN

4.2.5 Equivariant Transformer

4.2.6 Hamiltonian Neural Network

4.3 Matched Training Protocol

4.4 Output Accuracy

4.5 Equivariance Checks

CHAPTER FIVE

INTERPRETABILITY METHODOLOGY

[Defines every measurement instrument once: activation capture, linear probing with control baselines, causal ablation, activation patching, CKA (including the chosen definition of "matched depth" across heterogeneous models), and training-dynamics tracking.]

5.1 Activation Capture

5.2 Linear Probing

5.2.1 Probe Family and Metrics

5.2.2 Control Baselines

5.3 Causal Ablation

5.4 Activation Patching

5.5 Representational Similarity

5.6 Training-Dynamics Tracking

5.7 Compute and Reproducibility

CHAPTER SIX

RESULTS I: WHAT IS ENCODED

[Answers RQ1. Layer-by-layer physics encoding profiles for every architecture and ground-truth quantity, always shown against the control baselines; compares geometry (distance, direction) with law (force magnitude, potential).]

6.1 Physics Encoding Profiles

6.2 Geometry Versus Law

6.3 Depth of Emergence

6.4 The Hamiltonian Network Versus the Rest

6.5 Unconstrained Baselines

6.6 Summary of RQ1 Findings

CHAPTER SEVEN

RESULTS II: WHAT IS USED

[Answers RQ2. Ablation of probed directions and activation patching of a charge flip, separating quantities the models merely encode from quantities they causally use.]

7.1 Ablation Results

7.2 Correlate Versus Cause

7.3 Activation Patching: Tracing a Charge Flip

7.4 Encodes Versus Uses

7.5 Summary of RQ2 Findings

CHAPTER EIGHT

RESULTS III: REPRESENTATIONAL GEOMETRY AND DYNAMICS

[Answers RQ3 and RQ4 with cross-architecture CKA (plus a sensitivity check on the matched-depth definition), and RQ5 with probe accuracy over training. RQ5 is exploratory and may be folded into the Discussion.]

8.1 Cross-Architecture Similarity

8.2 Does the Implementation of Symmetry Matter?

8.3 Physics-Explicit Versus Symmetry-Constrained

8.4 Sensitivity to the Matched-Depth Definition

8.5 Generalization Dynamics

8.6 Summary of RQ3–RQ5 Findings

CHAPTER NINE

DISCUSSION

[Revisits the hypothesis against the results, draws implications for physics ML (representational alignment as a third evaluation axis) and for interpretability (what analytic ground truth buys), and states limitations.]

9.1 Revisiting the Central Hypothesis

9.2 Implications for Physics-Informed Machine Learning

9.3 Implications for Interpretability

9.4 Learned Law Versus Shortcut, Revisited

9.5 Limitations

CHAPTER TEN

CONCLUSION AND FUTURE WORK

[Restates the contributions with results in hand and lists future directions (other force laws, nonlinear probes, symbolic distillation, larger systems, transfer to molecular models).]

10.1 Summary of Contributions

10.2 Future Work

APPENDIX A

SIMULATOR AND VERIFICATION DETAILS

[Integrator derivation, softening choice, full verification tables.]

APPENDIX B

HYPERPARAMETERS AND TRAINING CURVES

[Per-model hyperparameter tables and all training curves.]

APPENDIX C

PROBE CONTROL BASELINES AND EXTENDED RESULTS

[Control-baseline plots that do not fit in Chapter 6; per-seed variance.]

APPENDIX D

ADDITIONAL REPRESENTATIONAL SIMILARITY ANALYSES

[CKA under alternative matched-depth definitions; linear vs. RBF kernels.]

APPENDIX E

CODE AND DATA AVAILABILITY

[Repository, software versions, commands to reproduce every figure.]

BIBLIOGRAPHY

[1] Víctor Garcia Satorras, Emiel Hoogeboom, and Max Welling. E(n) equivariant graph neural networks. In *Proceedings of the 38th International Conference on Machine Learning*. PMLR, 2021.

[2] Samuel Greydanus, Misko Dzamba, and Jason Yosinski. Hamiltonian neural networks. In *Advances in Neural Information Processing Systems*, 2019.

[3] Yi-Lun Liao and Tess Smidt. Equiformer: Equivariant graph attention transformer for 3d atomistic graphs. In *International Conference on Learning Representations*, 2023.

[4] Chris Olah, Nick Cammarata, Ludwig Schubert, Gabriel Goh, Michael Petrov, and Shan Carter. Zoom in: An introduction to circuits. *Distill*, 2020. doi: 10.23915/distill.00024.001.

[5] Neel Nanda, Lawrence Chan, Tom Liberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*, 2023.

[6] Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.

[7] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*. PMLR, 2019.

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